

Extension: Fitting functions to quadratic data



1

- | | | | | | |
|----------|--------------------|--------------------|----------|--------------------|--------------------|
| a | i Linear | ii Positive | b | i Linear | ii Negative |
| c | i Quadratic | ii Positive | d | i Quadratic | ii Negative |
| e | i Quadratic | ii Negative | f | i Quadratic | ii Negative |
| g | i Quadratic | ii Positive | h | i Quadratic | ii Negative |

2

- | | | | | | | |
|----------|---|----------|-------------|--------------|---------------|---------------|
| a | 1 | b | i No | ii No | iii No | iv Yes |
|----------|---|----------|-------------|--------------|---------------|---------------|

3

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|----------|--|----------|--|
| a | Quadratic, $f(x) = ax^2 + bx + c, a > 0$ | b | Quadratic, $f(x) = ax^2 + bx + c, a < 0$ |
|----------|--|----------|--|

4

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|----------|---|----------|--------------|--------------|---------------|--------------|
| a | 9 | b | i Yes | ii No | iii No | iv No |
|----------|---|----------|--------------|--------------|---------------|--------------|

Applications

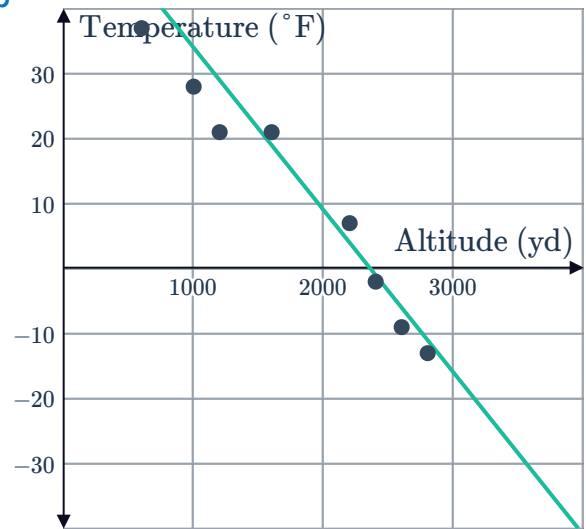
5

- | | | | |
|----------|------|----------|----|
| a | 67.3 | b | 35 |
|----------|------|----------|----|

6 **B**

7

a Linear function



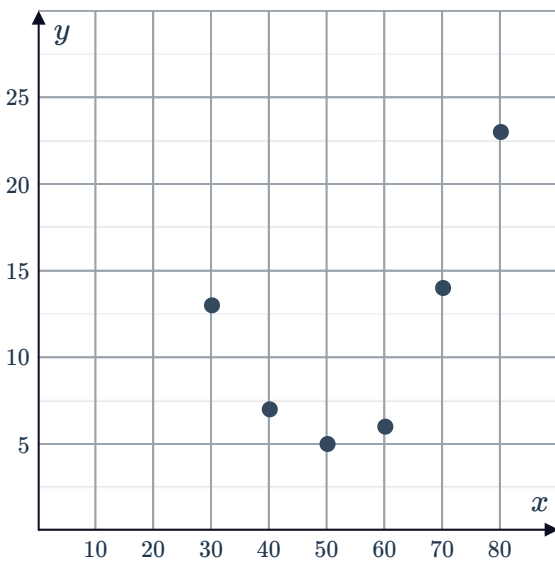
c

Altitude (x)	Average Annual Temperature (y)
500	46.5
1000	34
2000	9

d 36.25°F

8

a

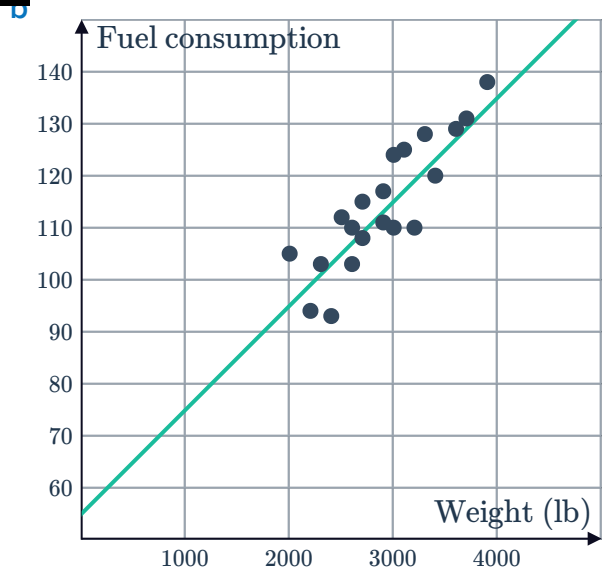


c $y = 0.02x^2 - 2.05x + 56.15$

b Quadratic

9

a Linear function



c

Weight (x)	Average Fuel Consumption (y)
3200	119
3100	117
1500	85

d The 1600 lb car

10

a

i 0.25 km

ii 4.375 km

b The extrapolation at $t = 8$ minutes is less reliable since the model predicts that Emma continues to increase her speed for the entire duration of her run.

11

a



c $y = ax^2 + bx + c$

d 6



b

No